INCH-POUND

MIL-B-49430/11F(ER)
28 July 1993
SUPERSEDING
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1 August 1991

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE, BA-5800/U

This specification sheet is approved for use within Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49430(ER).

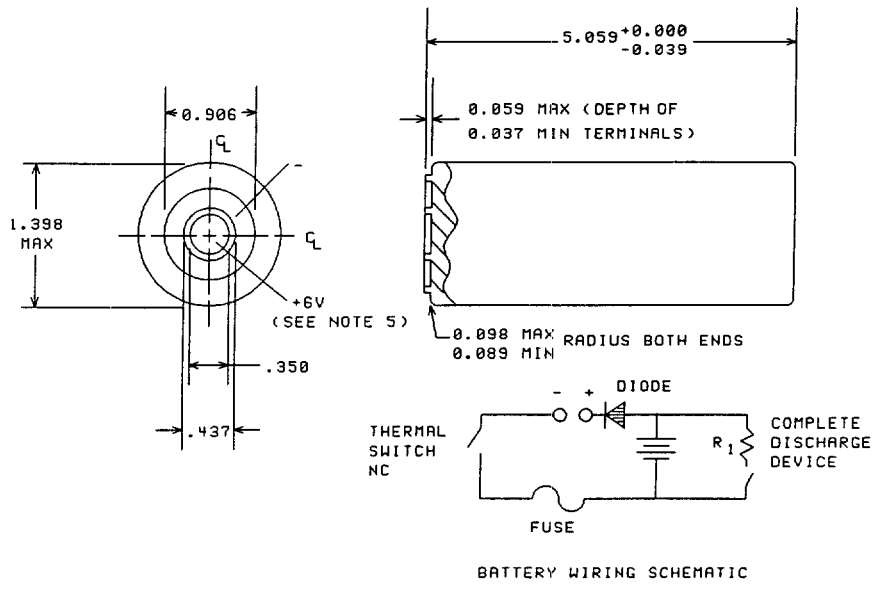


Figure 1. Battery dimensions and wiring schematic.

AMSC N/A FSC 6135

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NOTES:

1. All dimensions are in inches.

2. Tolerances are as follows (except as noted on Figure 1):

X.XXX ±0.004

3. Electrical terminal contacts are 0.01 inches thick.

4. R_1 value shall be between 60 ohms and 84 ohms and have a power rating of 0.5 watt.

5. Terminal markings to be printed or embossed on jacket in suitable location near the electrical contacts.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 6.10 volts.

Minimum voltage for capacity: 4.00 volts.

The battery shall contain a 2.25 amperes non-replaceable slow blow fuse wired in series with the cells in the negative leg.

The battery shall contain a diode wired in series. See paragraph 3.5.5.2 of basic specification.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery as described in paragraphs 3.5.5 and 3.5.5.1 of the basic specification. The high temperature switch may be connected at any place in the electrical circuit of the battery.

Weight (maximum): 0.48 pound (220 grams).

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.

Capacity tests: When the battery is tested in accordance with the method of examination and tests of this specification, the minimum capacity requirements shall be as specified below:

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Capacity Test	Battery Capacity Requirements in hours to 4.00 volts
I	3.2
L	1.6
H	6.5
HT	5.4
LT	1.4
T	2.8
D	2.8

Initial voltage delay: During the I, H, HT, T, and D discharge test, initial voltage below 4.00 volts cannot exceed a 5.0 second duration. During the L and LT discharge tests, initial voltages below 3.80 volts cannot exceed a 60.0 second duration.

METHODS OF EXAMINATION AND TESTS:

Capacity test:

(1) Storage: Details on storage conditions for all capacity tests are specified in the basic specification (see 4.8.9).

(2) Discharge:

Cells: Discharge at 2.0 amperes constant current at $70\pm5^{\circ}F$ for a minimum of 2.7 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 5.4 hours into voltage reversal.

Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts. There shall be no venting, fire or explosion when the batteries are discharged.

Closed-circuit voltage test: Load resistance of 5 ohms shall be used. Voltage shall be above 4.00 volts within 5 seconds.

Custodian:
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